Data Analytics - Mini Project

Supply Chain Analysis and Performance:

1.Project Overview:

The Shipment Performance Dashboard is a data analysis and visualization project developed using Microsoft Power BI. The project focuses on analyzing shipment data to understand shipping costs, shipment volumes, delivery performance, shipping modes, product categories, and shipping carriers.

The dataset contains information such as Shipment ID, Order Date, Supplier Name, Product Category, Quantity, Total Cost, Ship Date, Delivery Date, Shipping Carrier, Shipping Mode, and On-Time Delivery status.

The data was analyzed and transformed into meaningful measures and visualizations. The Power BI dashboard provides an interactive view of shipment performance and helps users easily identify trends, compare different categories, and monitor on-time and delayed shipments.

2.Project Objectives:

The main objectives of this project are:

1. To analyze the overall shipment performance using the available shipment data.
2. To calculate and understand the total shipping cost and total number of shipments.
3. To analyze the quantity of products shipped across different product categories and suppliers.
4. To compare shipping costs across different product categories.
5. To analyze on-time and delayed shipments and calculate the on-time delivery rate.
6. To identify shipping cost trends over different months and dates.
7. To compare shipping costs across different shipping modes such as Road, Sea, and Air.
8. To compare the number of shipments handled by different shipping carriers.
9. To provide interactive filters and slicers so users can analyze specific product categories, shipping modes, carriers, and date ranges.
10. To create an easy-to-understand dashboard that helps users make better decisions based on shipment performance.

3.Data Visualization:

The analysis was presented through an interactive Power BI dashboard using different types of visualizations.

1. KPI Cards

Cards are used to display important summary information:

- Total Shipping Cost
- Total Shipments
- On-Time Delivery Rate

These cards provide a quick overview of the overall shipment performance.

2. Bar Chart – Shipping Cost by Product Category

A bar chart is used to compare the total shipping cost for each product category. This helps identify which product categories contribute more to the overall shipping cost.

3. Donut Chart – On-Time vs Delayed Shipments

A donut chart shows the distribution of on-time and delayed shipments. It provides a clear view of the delivery performance.

4. Line Chart – Monthly Shipping Cost Trend

A line chart is used to show the change in shipping costs over time. Users can drill down from Year to Quarter and Month to examine the trend in more detail.

5. Column Chart – Shipping Cost by Shipping Mode

A column chart compares the total shipping cost for different shipping modes. This helps identify which shipping mode has a higher contribution to shipping costs.

6. Bar Chart – Number of Shipments by Carrier

A bar chart compares the number of shipments handled by each shipping carrier. This helps understand the shipment volume associated with different carriers.

7. Slicers

Interactive slicers are included for:

- Product Category

- Shipping Mode
- Shipping Carrier
- Order Date

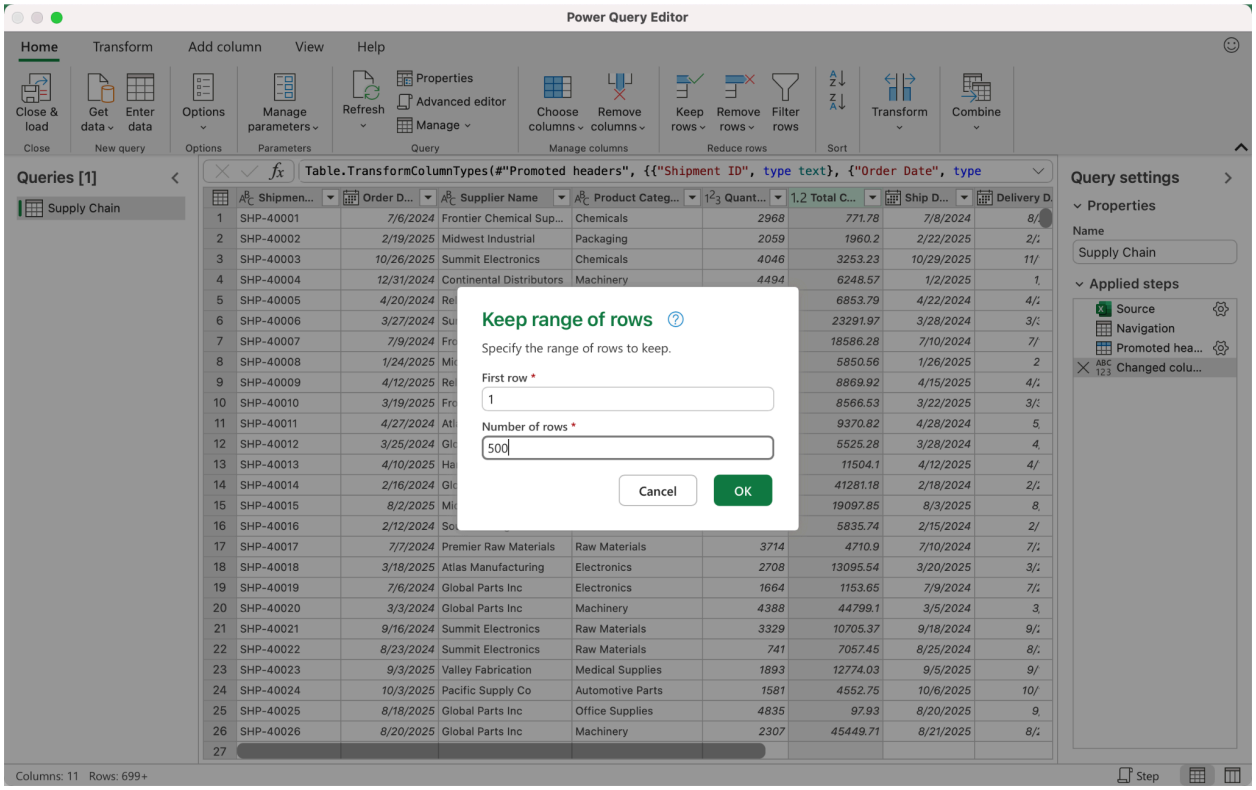
These slicers allow users to filter the dashboard and analyze specific parts of the shipment data.

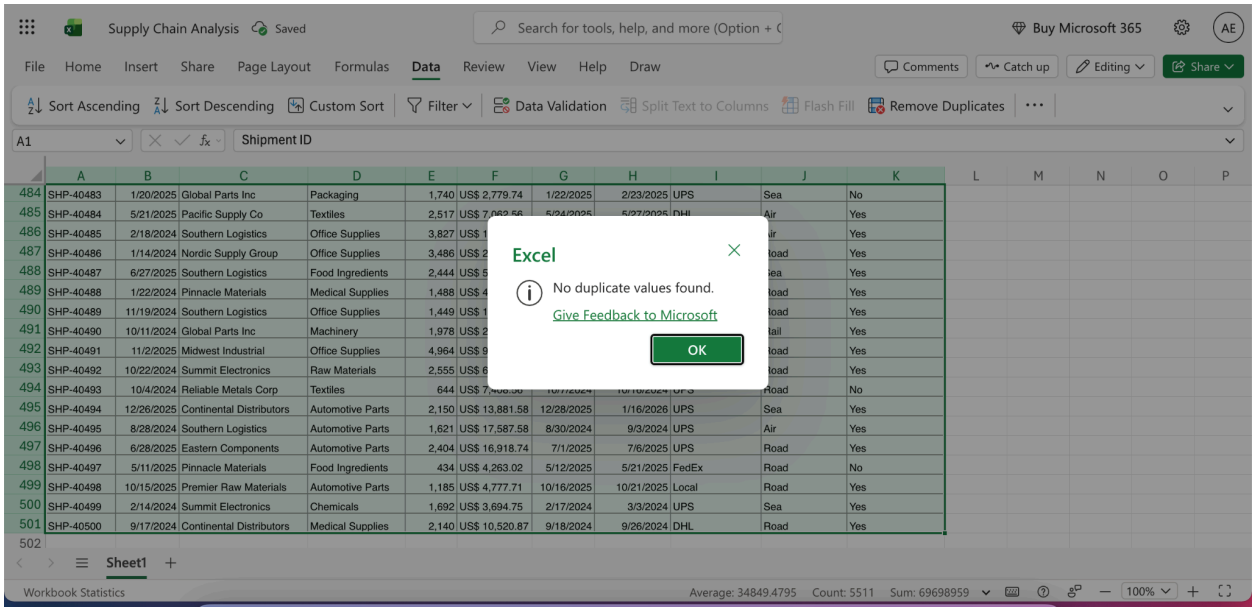
4.Conclusion:

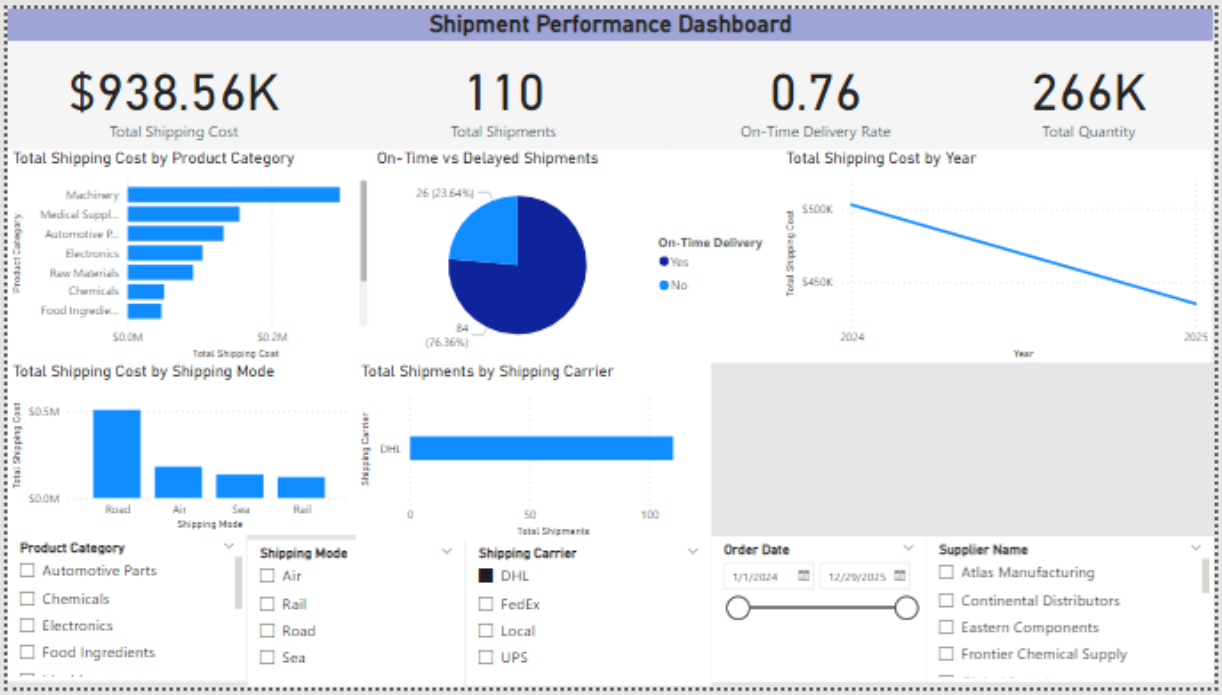
The Shipment Performance Dashboard successfully provides a clear and interactive view of shipment-related data using Microsoft Power BI. The analysis helps in understanding total shipping costs, shipment volumes, delivery performance, product categories, shipping modes, and shipping carriers.

The dashboard makes it easier to identify on-time and delayed shipments and observe shipping cost trends over time. The use of different visualizations such as cards, bar charts, column charts, donut charts, and line charts presents the information in a simple and understandable way.

Interactive slicers and drill-down features allow users to explore the data based on product category, shipping mode, shipping carrier, and order date. Overall, the dashboard transforms the shipment data into meaningful insights and provides a useful tool for monitoring shipment performance and supporting data-driven decision-making.


Power Query Editor
Keep range of rows
Specify the range of rows to keep.
First row *
1
Number of rows *
500
Cancel
OK
Queries [1]
Supply Chain
Query settings
Properties
Name
Supply Chain
Applied steps
Source
Navigation
Promoted hea...
Changed colu...
Table.TransformColumnTypes(#"Promoted headers", {{"Shipment ID", type text}, {"Order Date", type
Columns: 11 Rows: 699+


Supply Chain Analysis
Saved
Search for tools, help, and more (Option + G)
Buy Microsoft 365
File Home Insert Share Page Layout Formulas Data Review View Help Draw
Comments Catch up Editing Share
Sort Ascending Sort Descending Custom Sort Filter Data Validation Split Text to Columns Flash Fill Remove Duplicates
A1
Shipment ID
Excel
No duplicate values found.
Give Feedback to Microsoft
OK
Sheet1
Workbook Statistics
Average: 34849.4795 Count: 5511 Sum: 69698959
100%


Shipment Performance Dashboard
$938.56K
Total Shipping Cost
110
Total Shipments
0.76
On-Time Delivery Rate
266K
Total Quantity
Total Shipping Cost by Product Category
Machinery
Medical Suppl...
Automotive P...
Electronics
Raw Materials
Chemicals
Food Ingredie...
Product Category
$0.0M
$0.2M
Total Shipping Cost
On-Time vs Delayed Shipments
26 (23.64%)
84
(76.36%)
On-Time Delivery
Yes
No
Total Shipping Cost by Year
$500K
$450K
Total Shipping Cost
2024
2025
Year
Total Shipping Cost by Shipping Mode
$0.5M
$0.0M
Total Shipping Cost
Road
Air
Sea
Rail
Shipping Mode
Total Shipments by Shipping Carrier
DHL
Shipping Carrier
0
50
100
Total Shipments
Product Category
Automotive Parts
Chemicals
Electronics
Food Ingredients
Shipping Mode
Air
Rail
Road
Sea
Shipping Carrier
DHL
FedEx
Local
UPS
Order Date
1/1/2024
12/29/2025
Supplier Name
Atlas Manufacturing
Continental Distributors
Eastern Components
Frontier Chemical Supply